

ARXIV.ORG / ABS / 2504.03553 · AGENTIC AI · LLM PLANNING

KnowSelf: Agentic Knowledgeable Self-Awareness for LLM Agents

Teaching agents to autonomously decide **when to act directly**, **when to reflect**, and **when to consult external knowledge** — using only 15–26% of the knowledge that competing methods require.

Audience: ML Researchers · Graduate Students Task: Interactive Planning (ALFWorld) Backbone: Llama-8B · Gemma-2B · GPT-4o

Three Situational Modes: Fast · Slow · Knowledgeable Thinking



**Fast Thinking**
Agent directly produces the correct action with minimal deliberation
<ACT>

**Slow Thinking**
Agent self-corrects through reflection before committing to action
<REF>

**Knowledgeable Thinking**
Agent retrieves external knowledge — only when strictly necessary
<KNO>

KnowSelf Pipeline: Data Construction
→ **Two-Stage Training** → **Self-Aware Inference**



KnowSelf Heats a Mug & Places a Saltshaker

Where Strong Baselines Fail

ALFWorld Task A

"Put a hot mug in cabinet"

✗ OpenAI-01

Opens microwave → incorrectly attempts to **take an egg** from it. Fails to identify the mug-heating opportunity. **<WRONG ACTION>**

✓ KnowSelf (Llama-8B)

Generates **<REF>** — reflects that the open microwave should be used to **heat the mug**. Correctly completes the task.

ALFWorld Task B

"Put a saltshaker in drawer"

✗ DeepSeek-R1

Reflects that the drawer is open → tries to **close the drawer first** (irrelevant step). Action fails; task stuck. **<WRONG REFLECTION>**

✓ KnowSelf (Llama-8B)

Generates **<KNO>** — retrieves knowledge: *place the object without removing unrelated items*. Correctly places saltshaker in drawer.



84.33% on ALFWorld (Llama-8B) Using Only 15% Knowledge

Backbone	Method	Know%	Put	Clean	Heat	Cool	Exam.	Put 2	All ↑
GPT-4o (CLOSED-SOURCE FRONTIER)									
GPT-4o	ReAct	0%	83.3	74.2	69.6	85.7	77.8	64.7	76.1
GPT-4o	Reflexion	0%	100.0	87.1	73.9	90.5	83.3	70.6	85.1
GPT-4o	ExpeL	100%	95.8	83.9	69.6	81.0	88.9	52.9	79.9
LLAMA-8B (OPEN-SOURCE)									
Llama-8B	ReAct	0%	33.3	3.2	0.0	57.1	66.7	23.5	27.6
Llama-8B	Reflexion	0%	63.0	22.7	5.9	64.3	86.4	50.0	51.5
Llama-8B	ExpeL	100%	83.3	32.3	30.4	23.8	55.6	17.6	41.0
Llama-8B	ETO	0%	91.7	70.6	82.6	61.9	88.9	64.7	78.4
Llama-8B	KnowAgent	100%	87.5	93.5	65.2	66.7	61.1	64.7	75.4
Llama-8B	WKM	100%	95.8	87.1	87.0	61.9	66.7	52.9	77.6
Llama-8B	KnowSelf ★	15.0%	91.7	87.1	91.3	85.7	77.8	64.7	84.3
GEMMA-2B (COMPACT OPEN-SOURCE)									
Gemma-2B	ReAct	0%	0.0	9.7	0.0	4.8	44.4	0.0	9.0
Gemma-2B	ETO	0%	91.7	83.9	78.3	52.4	77.8	29.4	71.6
Gemma-2B	KnowAgent	100%	91.7	90.3	69.6	71.4	66.7	41.2	73.9
Gemma-2B	WKM	100%	91.7	87.1	78.3	71.4	61.1	52.9	76.1
Gemma-2B	KnowSelf ★	26.4%	87.5	93.5	73.9	76.2	83.3	52.9	79.9

Minimal Knowledge, Maximum Impact: -15-26% Usage vs. 100% for Competing Methods

KNOWLEDGE USAGE % (LOWER IS MORE EFFICIENT)



OVERALL ACCURACY ON ALFWORLD – ALL TASKS (%)



Self-Awareness Is the Missing Piece in Agent Planning

- 01 **Flood irrigation is harmful, not just wasteful.** Indiscriminate knowledge injection causes pattern collapse and suboptimal planning — 100%-knowledge methods consistently underperform KnowSelf.
- 02 **Situational self-awareness is learnable end-to-end.** Special-token classification via lightweight heuristic labeling on self-explored trajectories is sufficient — no human annotation or external supervision needed.
- 03 **KnowSelf (Llama-8B) outperforms GPT-4o baselines:** 84.3% vs GPT-4o ReAct (76.1%) and GPT-4o ExpeL (79.9%) on ALFWorld — with an open 8B model.

Dramatic efficiency advantage: KnowSelf uses 100% of its knowledge to plan, while GPT-4o methods use 100% of its knowledge to plan and 100% of its knowledge to execute.